

Section B

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2.

Artificial Intelligence : Frontiers of Machine
Intelligence and Reaping Benefits

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recently launched a software which can read and diagnose X-ray scans far better than a radiologist with 10+ years of experience. It works on the technology of Artificial Intelligence. This invention can potentially transform the face of healthcare all over the world. Not just healthcare, in words of Mr. Sundar Pichai (CEO of Google), Artificial Intelligence will have more dramatic impact than the invention

of electricity has had over the world.

This tremendous potential of the technology comes from the "brain" which the machines get. Artificial Intelligence is the ability of machines to not only acquire information but to process, analyze it and provide results on their own — without a need of human to constantly guide them. It is like teaching a child what is a cat. After learning, the child can detect a cat even from its tail, its voice and can even draw its picture. Machines of today, have acquired this intelligence so far only attributed to humans.

It all started with the first industrial revolution with inventions of machines like steam engine, printing press, railways etc. These machines replaced physical efforts of human labour. Later when Charles Babbage came up with the first computer, machines had entered an era where they were replacing mental efforts of humans. Today, we are at the frontier of machine intelligence where they can think for themselves.

we see this throughout our daily lives. Industries are using robots more and more to speed up their

very
good

production. Banks are using software to detect frauds. Governments are using it to control tax evasions. Growth of social media has been fuelled by the AI algorithms which decide what to show in our "news feeds". E-commerce websites are using it to better target their customers with advertisements. Countries are using the drones for the entire range of security (offensive & defensive) architecture. Thus, we are surrounded by the intelligent machines powered by A.I.

This has been at the frontier, at the pinnacle of all the

technological developments. New things are being invented and commercialised every day. At the current level, these machines possess Artificial Special Intelligence. They can perform specialised tasks such as analyzing X-rays, detecting financial frauds, recognising images etc. with more efficiency than humans. The next frontier is the Artificial General Intelligence, where they could completely act like humans, having special as well as general abilities. Amazon's Alexa and robot Sophia of Saudi Arabia are good examples in the direction.

AI is not just a technological

marvel. It could become a tool of transforming economy, society and the entire earth for better.

Industries can use it to boost their productivity. Employing machines in hazardous processes has the potential to save lakhs of human lives. Recently a boiler accident in NTPC plant in UP took over 7 lives. Such tragedies could easily be averted if robots were used for cleaning boiler furnaces. Apart from safety, use of machines also frees human labour from menial & repetitive tasks. This can provide them with required work-life balance while simultaneously preserving

full form

dignity of human labour.

AI can boost Agricultural

productivity and remuneration. Use of meteorological data for data processing could efficiently predict weather. Analysing market data could predict potential rates of various crops. These predictions could help in taking better sowing decisions. Precision irrigation systems, monitoring of weather & soil health could be powered by AI to utilise limited resources for maximum output

Healthcare sector, which

lacks good number of trained professionals could use software algorithms that detect diseases. This low cost diagnostic tools could

be spread even in remote regions so that healthcare access could be universalised as well as prioritised. Further, use of technology like Telemedicine could bring healthcare to the doorstep of people.

In the similar way, AI could augment Education sector by fulfilling need of expert educators. Education could be made personalised, accessible and affordable.

One major area where impact of AI is visible even today is the transportation sector. Companies like Tesla, Google, Volvo have launched driverless cars powered by AI. These cars could save lives lost in accidents (over 80%).

of them due to human errors) and save thousands of man-hours lost in traffic jams — almost always caused by irrational driving. Transportation being the arteries of the economy, this will have tremendous positive impact on economy.

Machines could help humans in pushing frontiers of research. It might become that cure for cancer, vaccine for malaria could be discovered by a machine.

Apart from this, in a security infrastructure they are becoming an indispensable tool.

Use of drones, bomb defusing robots etc. have the potential of saving lives of our soldiers in unnecessary warfare.

However, as with any other technology, AI brings its own set of challenges. Biggest challenge is to manage wide scale of unemployment due to increasing cases of machines replacing humans. World Bank estimates the job loss of over 50% of total jobs in coming 20 years. Along with job losses, there is a fear of rising inequalities. AI will be dividing the world into 2 parts - those which have this technology, and those who do not. Countries / companies having AI could attain unfair advantage and make this divide larger.

There are also multiple

ethical issues involved in use of AI. Whether it should be used to modify human body? whether it should be used to attack humans remotely (dilemma of just war)? whether a driverless car should save its driver or a pedestrian? On top of this, there is scenario of misuse of AI by fringe elements to disrupt normal human life.

Benefits of AI could only be reaped if could address these challenges. First and foremost, we need to address unemployment fears. We need to reskill people for jobs not taken over by machines. We need to train students for



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human skills like Creativity, critical thinking rather than rote learning.

The world needs to have an ethical framework under which to develop A.I. Cyber Geneva Convention proposed by Satya Nadella is needed. We need international forums like OpenAI (by Elon Musk) to think over potential good uses of AI and spread it across the globe.

There is no debate over use of AI. It is inevitable and necessary. It will bring a wave of progress for the entire humanity. Instead of drowning in it, we need to prepare our ships and sail towards the new world!